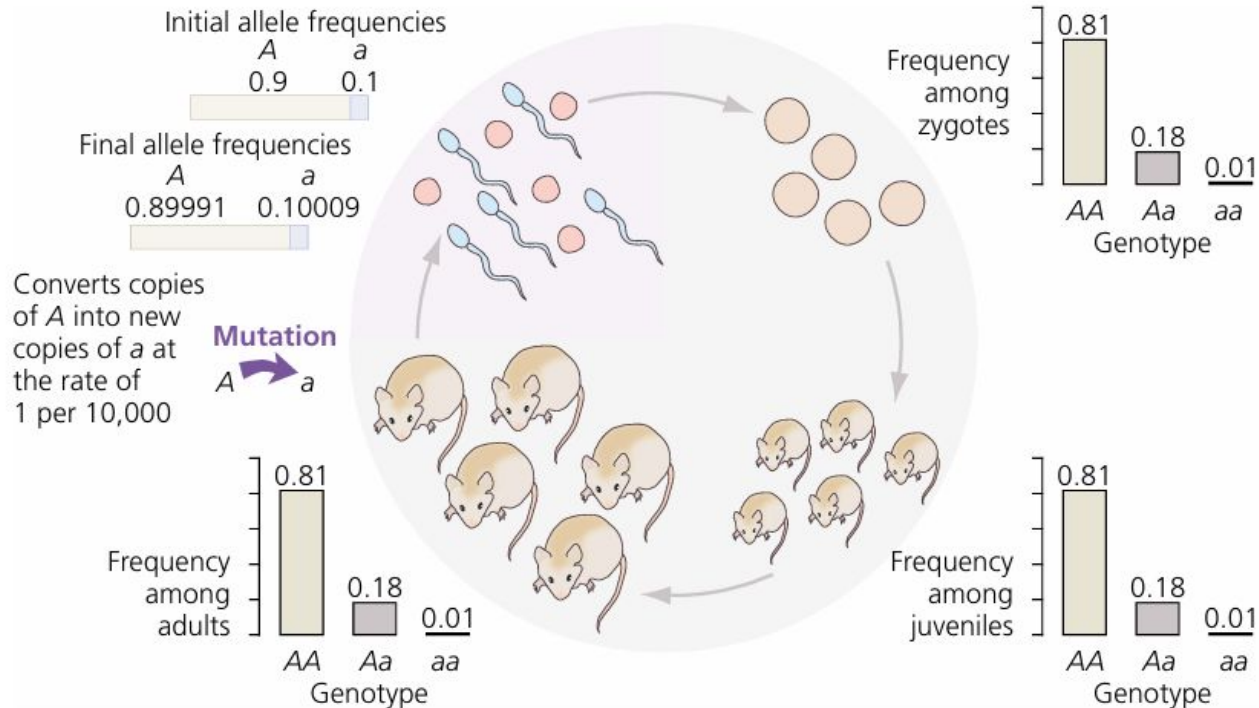


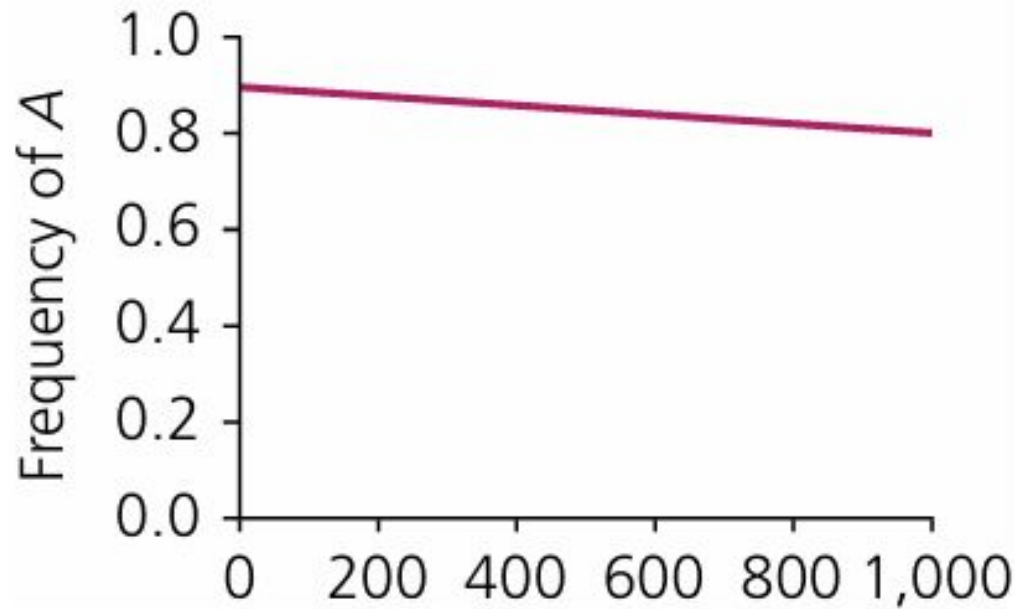
# BIO102 Help Session 7

Incorporating Mutation and Selection into the Hardy-Weinberg  
Equilibrium

# Adding Mutation to the Hardy–Weinberg Analysis: Mutation as an Evolutionary Mechanism



# Mutation is a Weak Mechanism of Evolution



# Mathematical Treatment

Imagine a single locus with two alleles: a wild-type allele,  $A$ , and a recessive loss-of-function mutation,  $a$ . Let  $\mu$  be the rate of mutation from  $A$  to  $a$ . Assume that the rate of back-mutation from  $a$  to  $A$  is negligible. If the frequency of  $A$  in this generation is  $p$ , then its frequency in the next generation is given by

$$p' = p - \mu p$$

If the frequency of  $a$  in this generation is  $q$ , then its frequency in the next generation is given by

$$q' = q + \mu p$$

The change in  $p$  from one generation to the next is

$$\Delta p = p' - p$$

which simplifies to

$$\Delta p = -\mu p$$

After  $n$  generations, the frequency of  $A$  is approximately

$$p_n = p_0 e^{-\mu n}$$

where  $p_n$  is the frequency of  $A$  in generation  $n$ ,  $p_0$  is the frequency of  $A$  in generation 0, and  $e$  is the base of the natural logarithms.

Readers familiar with calculus can derive the last equation as follows. First, assume that a single generation is an infinitesimal amount of time, so that we can rewrite the equation  $\Delta p = -\mu p$  as

$$\frac{dp}{dg} = -\mu p$$

Now divide both sides by  $p$ , and multiply both sides by  $dg$  to get

$$\left(\frac{1}{p}\right)dp = -\mu dg$$

Finally, integrate the left side from frequency  $p_0$  to  $p_n$  and the right side from generation 0 to  $n$ :

$$\int_{p_0}^{p_n} \left(\frac{1}{p}\right)dp = \int_0^n -\mu dg$$

and solve for  $p_n$ .

# Mutation-Selection Balance

Selection tends to eliminate such mutations from populations.

Deleterious alleles persist, however, because they are continually created anew.

When the rate at which copies of a deleterious allele are being eliminated by selection is exactly equal to the rate at which new copies are being created by mutation, the frequency of the allele is at equilibrium.

This situation is called mutation–selection balance

# Mutation-Selection Balance

**What is the frequency of the deleterious allele at equilibrium?**

Imagine a single locus with two alleles, A1 and A2, with frequencies  $p$  and  $q$ . A1 is the wild type; A2 is deleterious.

Let  $m$  be the rate at which copies of A1 are converted into copies of A2 by mutation.

Assume that the rate of back-mutation is negligible.

Selection continuously removes copies of A2 from the population, while mutation continuously creates new copies.

We want to calculate the frequency of A2 at which these processes cancel each other.

# Mutation-Selection Balance for a Deleterious Dominant

Suppose  $A$  is a dominant harmful mutation arising at a rate  $\mu$  and found at frequency  $p$

Selection acts against  $A$  at a value  $s$

| Genotype         | $aa$ | $Aa$    | $AA$    |
|------------------|------|---------|---------|
| Relative fitness | 1    | $1 - s$ | $1 - s$ |

# Mutation-Selection Balance for a Deleterious Dominant

Proportion of the allele 'a' in the population =  $1 - p$

Rate of creation of new mutants is  $\mu(1-p)$

Rate of loss of mutants is  $ps$

At equilibrium ( $p^*$ ), gain = loss

$$\mu(1-p^*) = p^*s$$

$$p^* \approx \mu / s$$

If the allele is dominant and lethal

$s=1$  and  $p^* = \mu$ .



# Mutation–selection balance for a deleterious recessive

Imagine that  $A_2$  is a deleterious recessive allele, such that the genotype fitnesses are given by

$$\begin{array}{ccc} w_{11} & w_{12} & w_{22} \\ 1 & 1 & 1 - s \end{array}$$

where the selection coefficient  $s$  gives the strength of selection against  $A_2$ .

First, we write an equation for  $p^*$ , the frequency of allele  $A_1$  after selection has acted, but before mutations occur. From Computing Consequences 6.3, this is

$$p^* = \frac{p^2 w_{11} + p q w_{12}}{p^2 w_{11} + 2 p q w_{12} + q^2 w_{22}}$$

Substituting the fitnesses from the table above, and  $(1 - p)$  for  $q$ , then simplifying gives

$$p^* = \frac{p}{1 - s(1 - p)^2}$$

Next we write an expression for  $p'$ , the frequency of allele  $A_1$  after mutations occur. Mutations convert a fraction  $\mu$  of the copies of  $A_1$  into copies of  $A_2$ , leaving behind a fraction  $(1 - \mu)$ . Thus

$$p' = (1 - \mu)p^* = \frac{(1 - \mu)p}{1 - s(1 - p)^2}$$

Finally, when mutation and selection are in balance,  $p'$  is equal to  $p$ , the frequency of allele  $A_1$  that we started with:

$$\frac{(1 - \mu)p}{1 - s(1 - p)^2} = p$$

This simplifies to

$$(1 - p)^2 = \frac{\mu}{s}$$

Substituting  $q$  for  $(1 - p)$  and solving for  $q$  yields an equation for  $\hat{q}$ , the equilibrium frequency of allele  $A_2$  under mutation–selection balance:

$$\hat{q} = \sqrt{\frac{\mu}{s}}$$

If  $A_2$  is a lethal recessive, then  $s = 1$ , and the equilibrium frequency of  $A_2$  is equal to the square root of the mutation rate.

**Example 1a:**

A population of cats can be either black or white; the black allele (B) has complete dominance over the white allele (b). Given a population of 1,000 cats, 840 black and 160 white, determine the allele frequency, the frequency of individuals per genotype, and number of individuals per genotype.

To solve this problem, solve for all the preceding variables ( $p$ ,  $q$ ,  $p^2$ ,  $2pq$ , and  $q^2$ ).

**Step 1:** Find the frequency of white cats, the homozygous recessive genotype, as they have only one genotype, bb. Black cats can have either the genotype Bb or the genotype BB, and therefore, the frequency cannot be directly determined.

$$\text{Frequency of individuals} = \frac{\text{Individuals}}{\text{Total Population}}$$

$$\frac{\text{Individuals}}{\text{Total Population}} = \frac{160}{1,000} = 0.16$$

Frequency of white cats = 0.16; therefore,  $q^2 = 0.16$

**Step 2:** Find  $q$  by taking the square root of  $q^2$ .

$$\begin{aligned}\sqrt{(q^2)} &= \sqrt{(0.16)} \\ q &= 0.4\end{aligned}$$

**Step 3:** Use the first Hardy-Weinberg equation ( $p + q = 1$ ) to solve for  $p$ .

$$\begin{aligned}p + q &= 1 \\ p &= 1 - q \\ p &= 1 - (0.4) \\ p &= 0.6\end{aligned}$$

Now that the allele frequencies in the population are known, solve for the remaining frequency of individuals by using  $p^2 + 2pq + q^2 = 1$ .

**Step 4:** Square  $p$  to find  $p^2$ .

$$\begin{aligned}p &= 0.6 \\p^2 &= (0.6)^2 \\p^2 &= 0.36\end{aligned}$$

**Step 5:** Multiply  $2 \times p \times q$  to get  $2pq$ .

$$\begin{aligned}2pq &= 2(0.6)(0.4) \\2pq &= 0.48\end{aligned}$$

**Therefore:**

The frequency of the dominant alleles:  $p = 0.6$

The frequency of the recessive alleles:  $q = 0.4$

The frequency of individuals with the dominant genotype:  $p^2 = 0.36$

The frequency of individuals with the heterozygous genotype:  $2pq = 0.48$

The frequency of individuals with the recessive genotype:  $q^2 = 0.16$

**Remember:** Frequencies can be checked by substituting the values above back into the Hardy-Weinberg equations.

$$0.6 + 0.4 = 1$$

$$0.36 + 0.48 + 0.16 = 1$$

**Step 6:** Multiply the frequency of individuals ( $p^2$ ,  $2pq$ , and  $q^2$ ) by the total population to get the number of individuals with that given genotype.

$$p^2 \times \text{total population} = 0.36 \times 1,000 = 360 \text{ black cats, BB genotype.}$$

$$2pq \times \text{total population} = 0.48 \times 1,000 = 480 \text{ black cats, Bb genotype.}$$

$$q^2 \times \text{total population} = 0.16 \times 1,000 = 160 \text{ white cats, bb genotype.}$$



### Comparing Generations

To know if a population is in Hardy-Weinberg Equilibrium scientists have to observe *at least two* generations. If the allele frequencies are the same for both generations then the population is in Hardy-Weinberg Equilibrium.

#### Example 1b:

Recall: the previous generation had allele frequencies of  $p = 0.6$  and  $q = 0.4$ .

The next generation of cats has a total population of 800 cats, 672 black and 128 white. Is the population in Hardy-Weinberg Equilibrium?

**Step 1:** Solve for  $q^2$ .

$$\frac{\text{Individuals with the Recessive Genotype}}{\text{Total Population}} = \frac{128}{800} = 0.16$$

**Step 2:** Use  $q^2$  to solve for  $q$ . There is no need to solve the entire equation, because if  $q$  has changed, then  $p$  has also changed. If  $q$  remains the same, then  $p$  will remain the same.

$$\begin{aligned}q^2 &= 0.16 \\ \sqrt{q^2} &= \sqrt{0.16} \\ q &= 0.4\end{aligned}$$

Because the recessive allele frequency ( $q$ ) has remained the same, the population is in a state of Hardy-Weinberg Equilibrium.

We start with a hypothetical population where frequency of A allele is 0.6 and frequency of a allele is 0.4, and assume that the AA and Aa has relative fitness 1 and aa homozygote has a relative fitness of 0.1 (i.e. 90% of the aa offspring die without reproducing). What is the frequencies of A and a alleles would be in F1 generation?

$$\begin{array}{l}
 \begin{array}{c} \text{AA} \quad \text{Aa} \quad \text{aa} \\ F = p^2 + 2pq + q^2 = 1 \\ \text{aa} \quad 0,36 + 0,48 + 0,16 = 1 \\ q^2 = 0,4^2 = 0,16 \end{array} \quad \begin{array}{c} \text{A} \quad \text{a} \\ p + q = 1 \\ \underline{0,6} + \underline{0,4} = 1 \end{array} \\
 \begin{array}{c} \text{AA} \\ p^2 = 0,6^2 = 0,36 \\ \text{Aa} \\ 2pq = 2 \times 0,6 \times 0,4 = 0,48 \end{array} \\
 F = p^2 (w_{\text{AA}}) + 2pq (w_{\text{Aa}}) + q^2 (w_{\text{aa}}) \\
 F = \underline{0,36} (1) + \underline{0,48} (1) + \underbrace{0,16 \times 0,1}_{0,016} = 0,856
 \end{array}$$



We start with a hypothetical population where frequency of A allele is 0.6 and frequency of a allele is 0.4, and assume that the AA and Aa has relative fitness 1 and aa homozygote has a relative fitness of 0.1 (i.e. 90% of the aa offspring die without reproducing). What is the frequencies of A and a alleles would be in F1 generation?

$$\begin{array}{lll}
 \text{AA} & \text{Aa} & \text{aa} \\
 F = \frac{0,36}{0,856} = 0,42 & F = \frac{0,48}{0,856} = 0,56 & F = \frac{0,016}{0,856} = 0,019
 \end{array}$$

$$\begin{array}{c}
 A \quad a \\
 p + q = 1
 \end{array}$$

$$\begin{array}{ll}
 A & a \\
 p = 0,42 + \frac{1}{2} 0,56 = 0,7 & q = 1 - 0,7 = 0,3
 \end{array}$$

Nick Waser and Mary Price (1981. *Evolution* 35:376-90) examined selection against albino forms of the Rocky Mountain wildflower *Delphinium nelsonii*. They showed that the white-flowered albino plants received fewer visits from pollinators and produced fewer seeds per flower than the normal blue-flowered plants. Nevertheless, the rare white flowered plants were present in most populations. They suggested that those variants might be maintained by mutation selection balance.



Here are some of their data:

|                | Frequency of white-flowered plants |
|----------------|------------------------------------|
| site1          | 0.00096                            |
| site2          | 0.00054                            |
| site3          | 0.00103                            |
| site4          | 0.00042                            |
| <b>Average</b> | <b>0.00074</b>                     |

|                       | Average seed set (# seeds per flower) | Relative fitness |
|-----------------------|---------------------------------------|------------------|
| Blue flowered plants  | 19.7                                  | 1.0              |
| White flowered plants | 15.8                                  | 0.80             |

- If the relative fitness of albinos is 0.8, what is their value for  $s$ ?
- Assuming HWE and that the albino allele is completely recessive, what is the observed allele frequency,  $q$ ?
- What mutation rate would be required for the white allele to be maintained by mutation/selection balance?

The fitness of the recessive homozygote is defined as  $w=1-s$  , so if  $w=0.8$ , then  $s=0.2$  The expected frequency of white flowered plants is  $q^2$ , so  $q=\sqrt{0.00074} = 0.0272$

The mutation rate would have to be 0.00015